

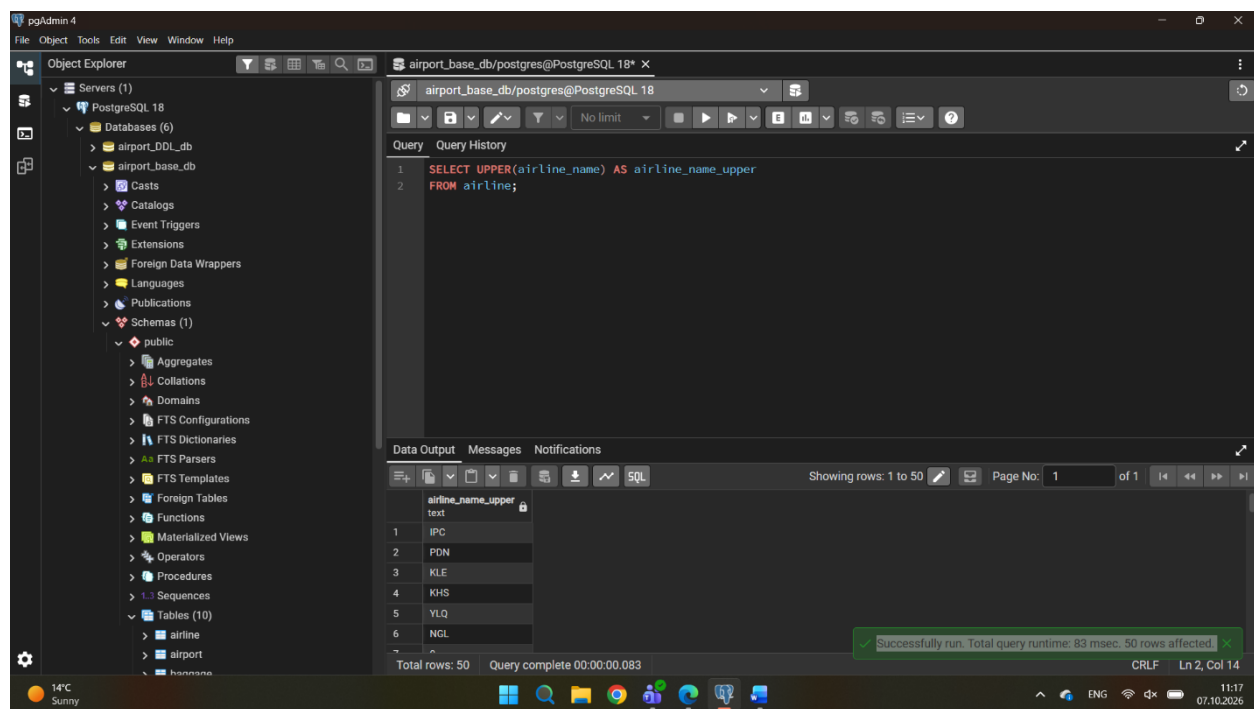
Laboratory work 4

We continue to work with the database from the previous laboratory works.

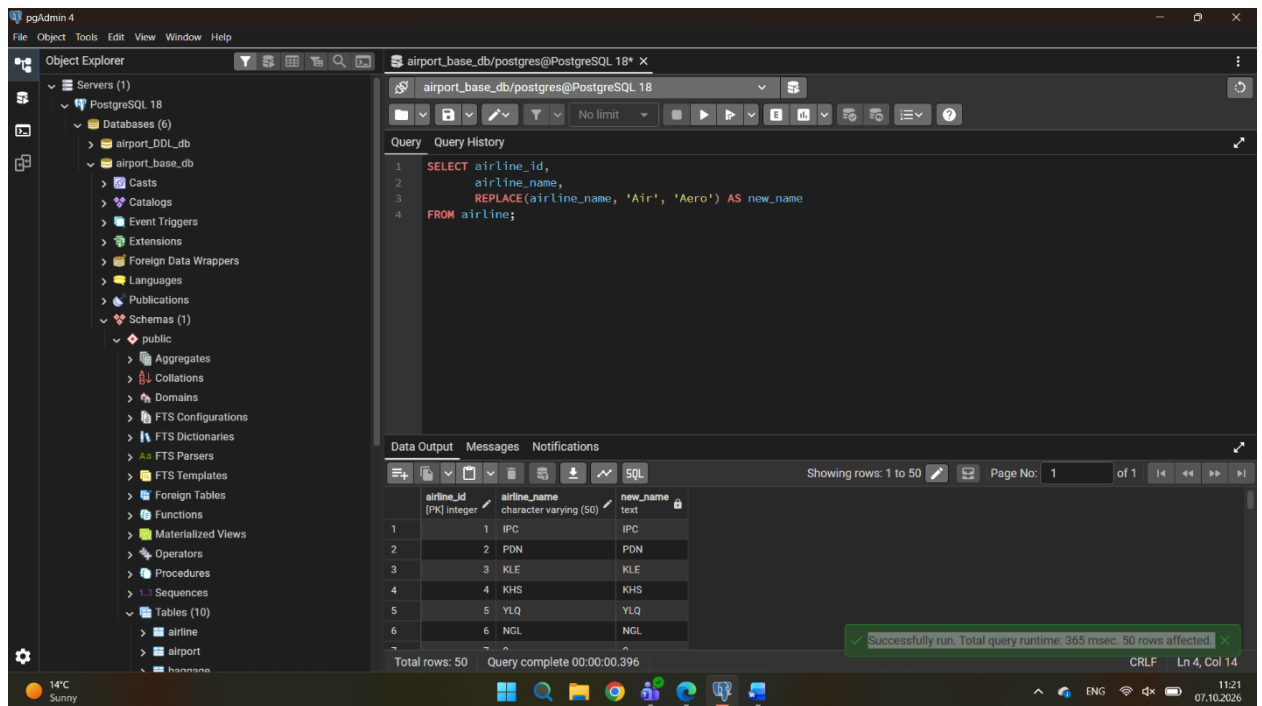
Take a full-page screenshot that covers the code and results of each task.

Tasks:

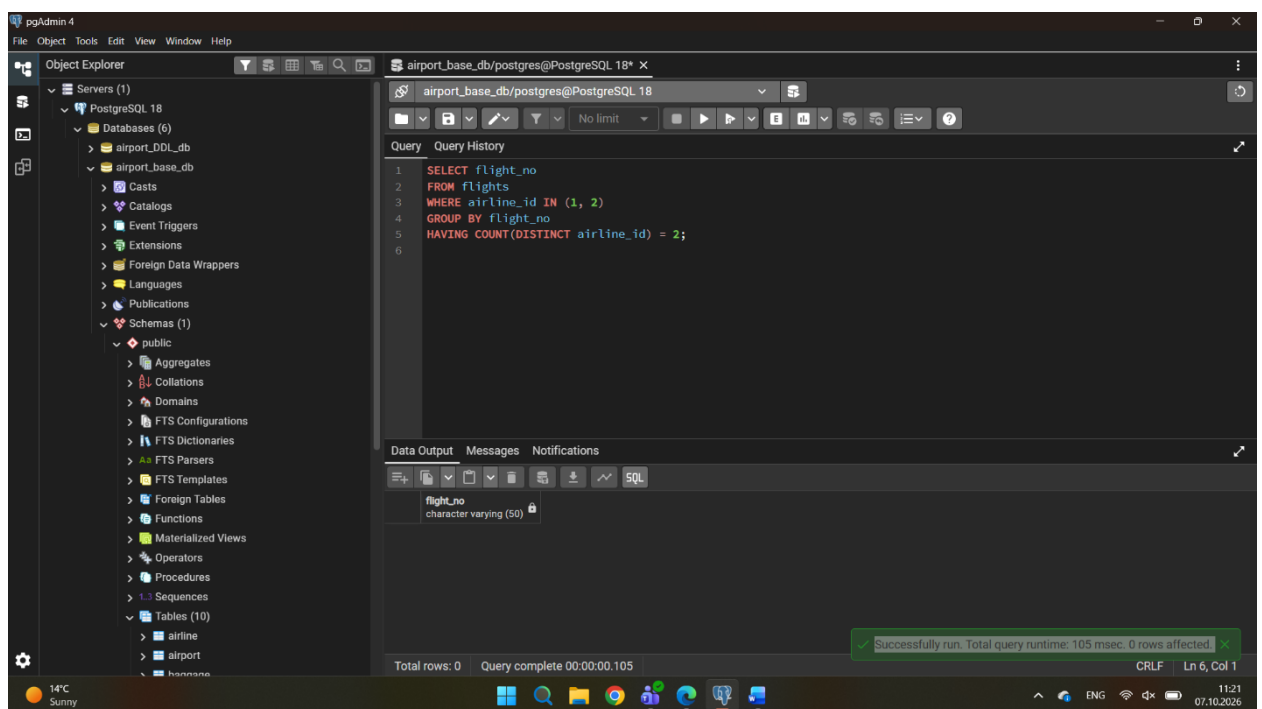
1. Retrieve all airline names in uppercase.



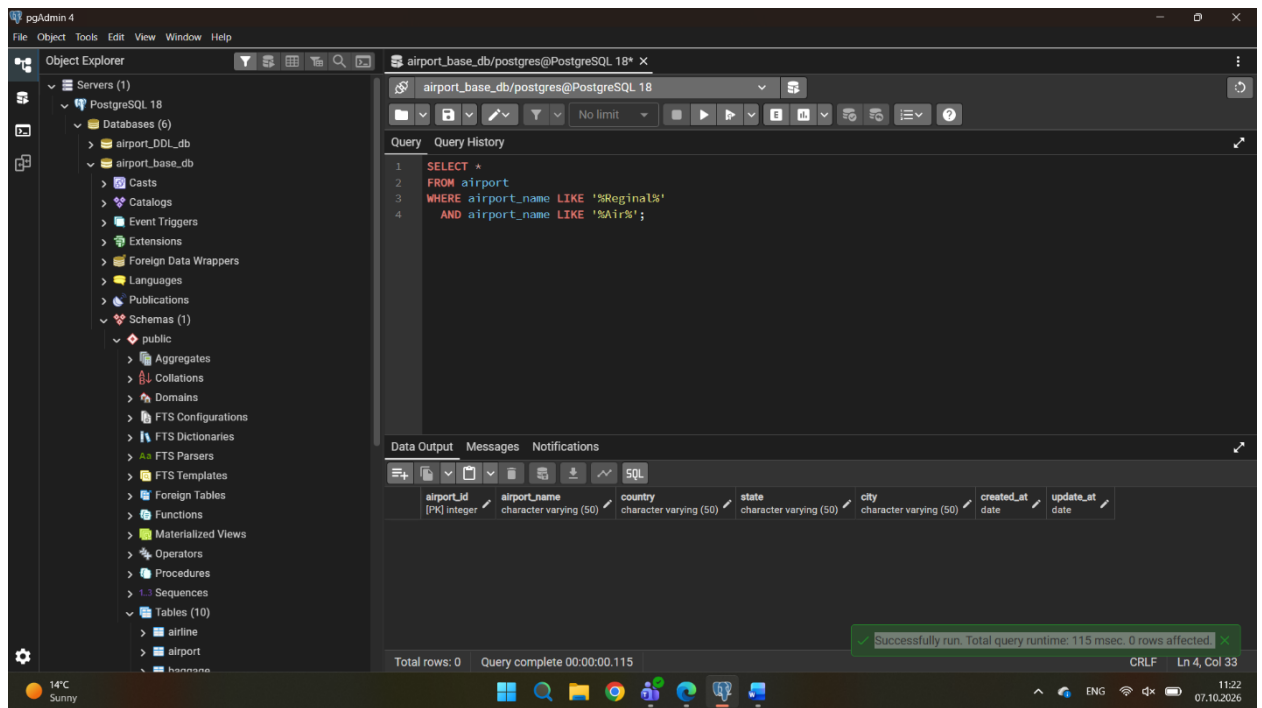
2. Replace any occurrence of the word "Air" in airline names with "Aero".



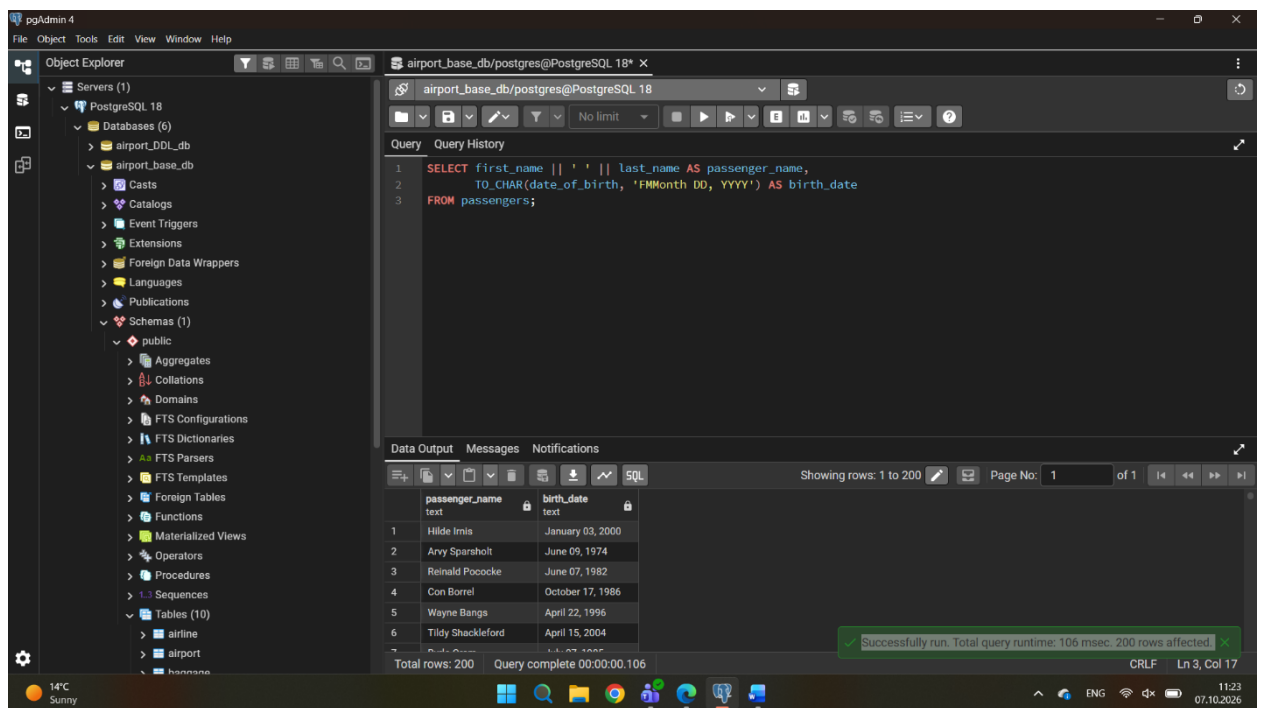
3. Find all flight numbers that coordinates with both airline 1 and airline 2.



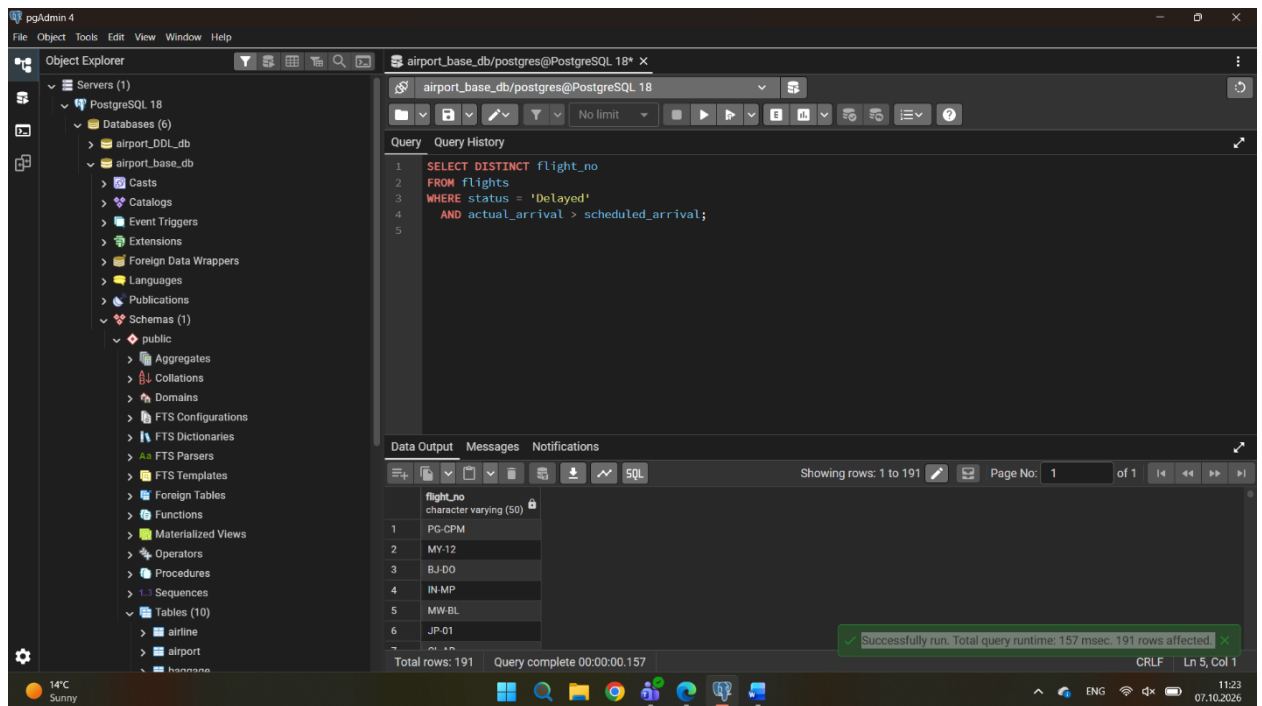
4. Retrieve airports that contain the word "Reginal" and "Air" in their names.



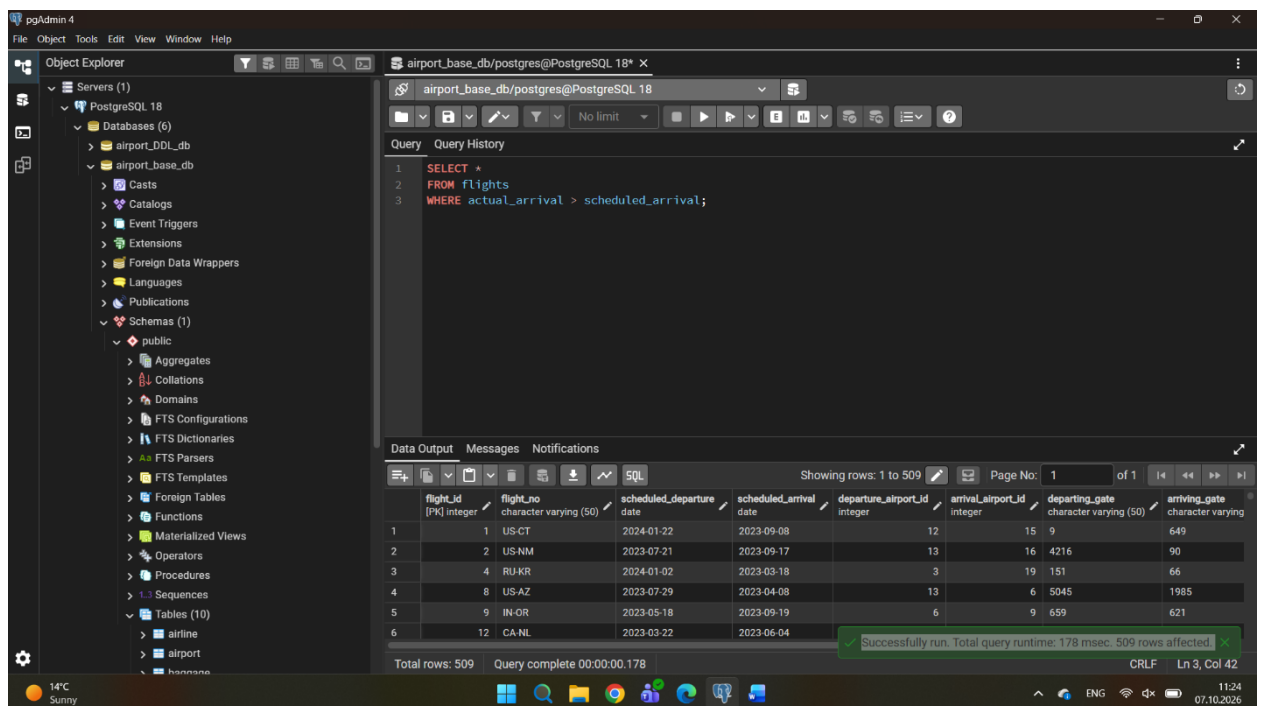
5. Retrieve passenger names and format their birth dates as 'Month DD, YYYY'.



6. Find flight numbers that have been delayed based on the actual arrival time.



7. Find flights that arrived late according to their actual arrival time compared to the scheduled arrival time.



8. Show airlines from any of: ('France','Portugal','Poland') created between '2023-11-01' and '2024-03-31'.

The screenshot shows the pgAdmin 4 interface with the 'airport_base_db' database selected. The 'airline' table is expanded in the Object Explorer. The SQL query editor contains the following query:

```

1 SELECT *
2 FROM airline
3 WHERE airline_country IN ('France', 'Portugal', 'Poland')
4 AND created_at BETWEEN '2023-11-01' AND '2024-03-31';

```

The Data Output tab shows the results of the query, displaying 7 rows. The columns are: airline_id [PK] integer, airline_code character varying (50), airline_name character varying (50), airline_country character varying (50), created_at date, and update_at date.

airline_id [PK] integer	airline_code character varying (50)	airline_name character varying (50)	airline_country character varying (50)	created_at date	update_at date
4	OOKB	KHS	France	2023-11-18	2024-02-18
13	OIFE	IFH	Portugal	2024-01-09	2023-11-09
15	SYKZ	KKG	Poland	2023-12-24	2023-10-30
24	ZSRG	RUG	Portugal	2024-03-11	2023-08-24
28	RKPD	JDG	Poland	2023-11-03	2023-12-06
30	CYST	YST	Poland		
31			France		

Total rows: 7. Query complete 00:00:00.143. Successfully run. Total query runtime: 143 msec. 7 rows affected.

9. Find top3 overweighted baggage with more than 25kg.

The screenshot shows the pgAdmin 4 interface with the 'airport_base_db' database selected. The 'baggage' table is expanded in the Object Explorer. The SQL query editor contains the following query:

```

1 SELECT *
2 FROM baggage
3 WHERE weight_in_kg > 25
4 ORDER BY weight_in_kg DESC
5 LIMIT 3;

```

The Data Output tab shows the results of the query, displaying 3 rows. The columns are: baggage_id [PK] integer, weight_in_kg numeric (4,2), created_date date, update_date date, and booking_id integer.

baggage_id [PK] integer	weight_in_kg numeric (4,2)	created_date date	update_date date	booking_id integer
321	49.80	2023-07-07	2023-11-10	349
61	49.57	2023-04-21	2023-07-31	464
102	49.44	2024-01-24	2023-11-01	53

Total rows: 3. Query complete 00:00:00.130. Successfully run. Total query runtime: 130 msec. 3 rows affected.

10. Find the youngest passengers' full name.

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure for 'airport_base_db'. The central pane contains a SQL query:

```

1 SELECT *
2 FROM baggage
3 WHERE weight_in_kg > 25
4 ORDER BY weight_in_kg DESC
5 LIMIT 3;
6

```

The 'Data Output' pane shows the results of the query:

baggage_id [PK] integer	weight_in_kg numeric (4,2)	created_date date	update_date date	booking_id integer
321	49.80	2023-07-07	2023-11-10	349
61	49.57	2023-04-21	2023-07-31	464
102	49.44	2024-01-24	2023-11-01	53

A status message at the bottom indicates: 'Successfully run. Total query runtime: 130 msec. 3 rows affected.'

11. Find the cheapest booking price on each booking platform.

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure for 'airport_base_db'. The central pane contains a SQL query:

```

1 SELECT booking_platform,
2        MIN(price) AS cheapest_price
3 FROM booking
4 GROUP BY booking_platform;

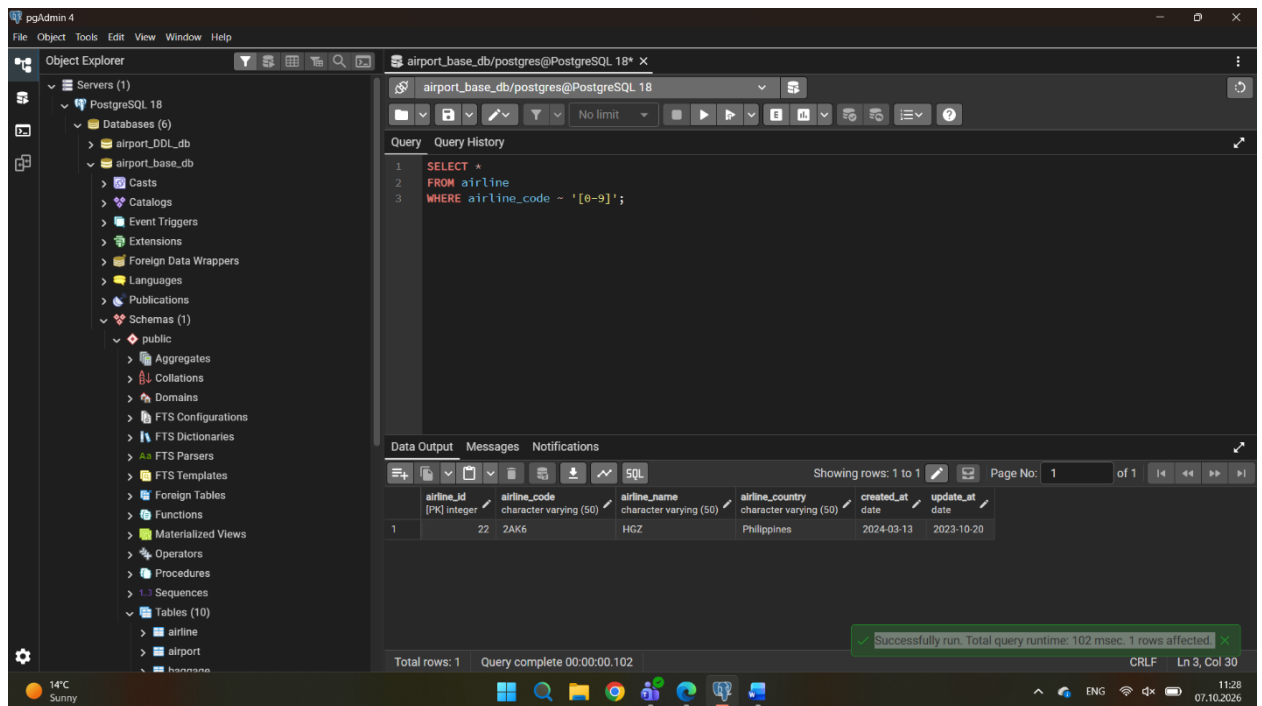
```

The 'Data Output' pane shows the results of the query:

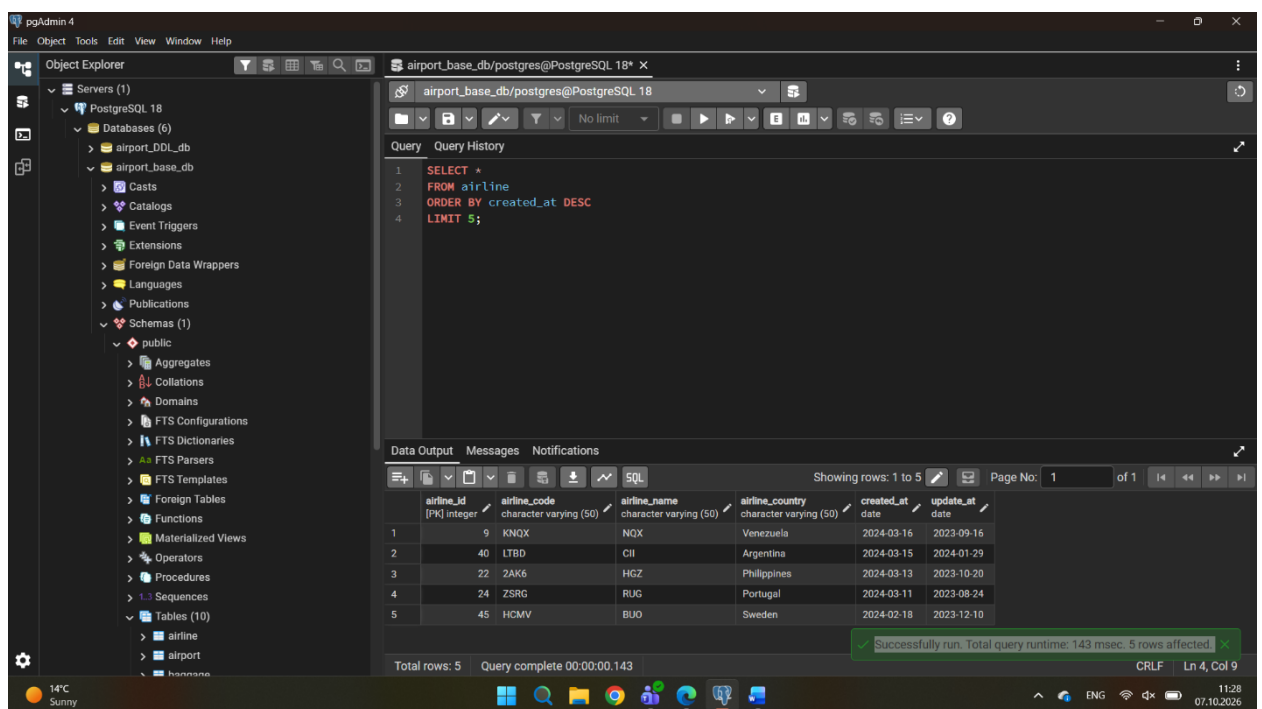
booking_platform character varying (50)	cheapest_price numeric
Schmitt-Quigley	4966.58
Rohan, Hamill and Bosco	7118.84
Kutch-Russel	2113.46
Veum-Corwin	2183.14
Wolff, Wuckert and Satterfield	4911.89
Reichel, Deckow and Quitzon	8389.34

A status message at the bottom indicates: 'Successfully run. Total query runtime: 164 msec. 494 rows affected.'

12. Return airlines whose airline_code contains a digit.



13. List the top 5 most recently created airlines.



14. Return all rows where booking_id is between 200 and 300 inclusive and check_result <> 'Checked'.

pgAdmin 4

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Object Explorer

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Tables (10)

airline

airport

baggage

Query

Query History

```

1 SELECT *
2 FROM baggage_check
3 WHERE booking_id BETWEEN 200 AND 300
4 AND check_result <> 'Checked';

```

Data Output Messages Notifications

Showing rows: 1 to 56 Page No: 1 of 1

baggage_check_id [PK] integer	check_result character varying (50)	created_at date	update_at date	booking_id integer	passenger_id integer	
1	7	Not checked	2023-05-04	2023-04-29	226	44
2	12	Not checked	2023-10-02	2023-06-22	288	133
3	22	Not checked	2023-06-14	2023-11-10	246	101
4	23	Not checked	2023-05-29	2023-06-02	284	22
5	24	Not checked	2024-02-17	2023-10-03	267	68
6	25	Not checked	2024-01-20	2023-06-10	23	

Total rows: 56 Query complete 00:00:00.167

Successfully run. Total query runtime: 167 msec. 56 rows affected.

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15. Baggage checks where update_at is in the same month as created_at but occurs earlier than created_at.

pgAdmin 4

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airline

airport

baggage

Query

Query History

```

1 SELECT *
2 FROM baggage_check
3 WHERE DATE_TRUNC('month', update_at) = DATE_TRUNC('month', created_at)
4 AND update_at < created_at;

```

Data Output Messages Notifications

Showing rows: 1 to 18 Page No: 1 of 1

baggage_check_id [PK] integer	check_result character varying (50)	created_at date	update_at date	booking_id integer	passenger_id integer	
1	34	Checked	2023-04-25	2023-04-08	28	57
2	86	Checked	2023-12-25	2023-12-07	312	22
3	104	Checked	2024-03-17	2024-03-14	41	83
4	109	Not checked	2023-08-30	2023-08-09	469	91
5	119	Not checked	2023-12-02	2023-12-01	135	55
6	148	Checked	2024-03-17	2024-03-16	8	

Total rows: 18 Query complete 00:00:00.120

Successfully run. Total query runtime: 120 msec. 18 rows affected.

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